

SET 7: EMPLOYEE PERFORMANCE ANALYSIS

Dataset: Employee Performance

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

PROGRAM 1: Line Chart of Employee Performance

Experiment: Data Visualization using R

Question: Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

Aim

To create a line chart in R that shows the performance trend of employees over their years of service, with the axes labelled and a title.

Dataset

The Employee Performance dataset given above is used. Years of Service is used as the time measure on the X-axis.

Procedure

1. Create the Employee Performance data frame in R.
2. Arrange the data in increasing order of Years of Service using `order()`.
3. Use `plot()` with `type = "o"` to draw the line chart with points, using Years of Service on the X-axis and Performance Score on the Y-axis.
4. Add the title, X-axis label and Y-axis label inside `plot()`.
5. Use `axis()` to show the years of service on the X-axis and `text()` to display the score at each point.

```
# Dataset
employees <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

# Arrange the data in order of Years of Service
```

```

employees <- employees[order(employees$Years_of_Service), ]

# Line Chart
plot(employees$Years_of_Service, employees$Performance_Score,
     type = "o",
     pch = 19,
     col = "blue",
     main = "Employee Performance Trend",
     xlab = "Years of Service",
     ylab = "Performance Score",
     ylim = c(70, 100),
     xaxt = "n")

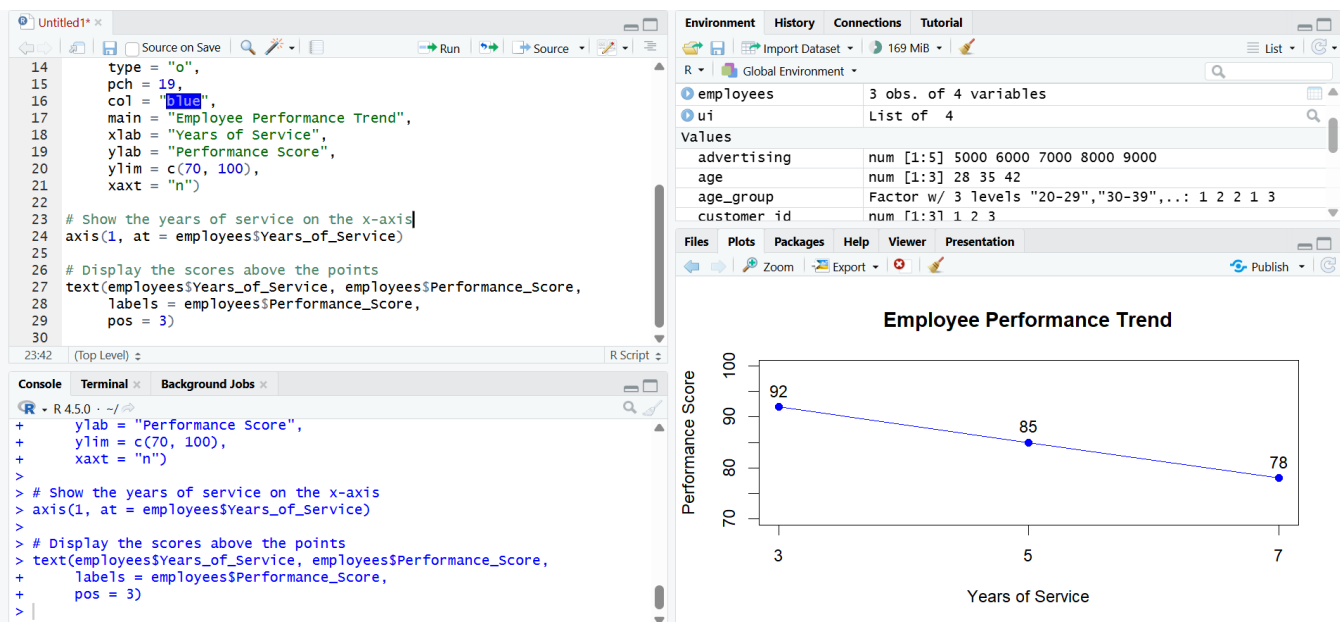
# Show the years of service on the x-axis
axis(1, at = employees$Years_of_Service)

# Display the scores above the points
text(employees$Years_of_Service, employees$Performance_Score,
     labels = employees$Performance_Score,
     pos = 3)

```

Expected Output

A line chart titled Employee Performance Trend is displayed with Years of Service on the X-axis and Performance Score on the Y-axis. The line passes through 92 at 3 years, 85 at 5 years and 78 at 7 years of service, so the performance score decreases as the years of service increase in this dataset.



Output 1: Employee Performance Trend (Line Chart)

Result

The line chart was successfully created in R to visualize the performance trend of employees, with the title, X-axis label and Y-axis label.

PROGRAM 2: Bar Chart of Employees by Department

Experiment: Data Visualization using R

Question: Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

Aim

To create a bar chart in R that shows the distribution of employees across the departments, with the chart elements labelled.

Procedure

1. Create the Employee Performance data frame in R.
2. Count the number of employees in each department using table().
3. Use barplot() to draw the bar chart of the counts, with the Department on the X-axis and the Number of Employees on the Y-axis.
4. Add the title, X-axis label and Y-axis label inside barplot().
5. Use axis() to show whole numbers on the Y-axis and text() to display the count on top of each bar.

```
# Dataset
employees <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

# Count the number of employees in each department
dept_count <- table(employees$Department)

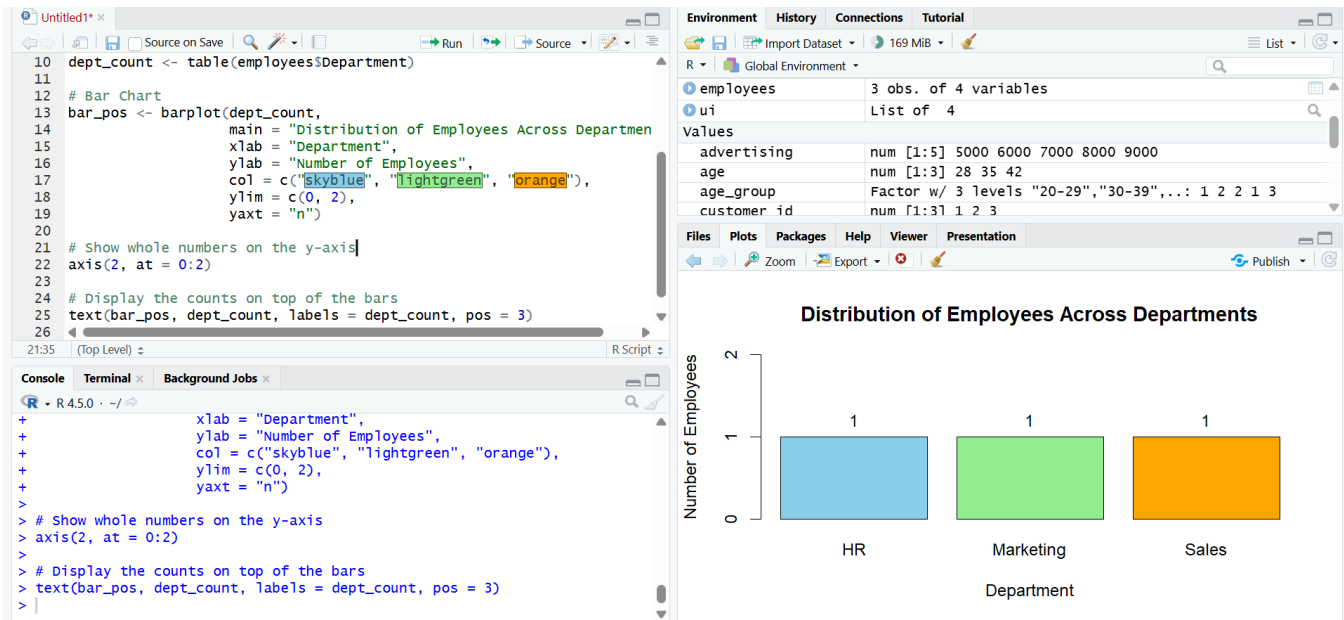
# Bar Chart
bar_pos <- barplot(dept_count,
  main = "Distribution of Employees Across Departments",
  xlab = "Department",
  ylab = "Number of Employees",
  col = c("skyblue", "lightgreen", "orange"),
  ylim = c(0, 2),
  yaxt = "n")

# Show whole numbers on the y-axis
axis(2, at = 0:2)

# Display the counts on top of the bars
text(bar_pos, dept_count, labels = dept_count, pos = 3)
```

Expected Output

A bar chart titled Distribution of Employees Across Departments is displayed with Department on the X-axis and Number of Employees on the Y-axis. HR, Marketing and Sales each have one employee, so all three bars have the same height of 1.



Output 2: Distribution of Employees Across Departments (Bar Chart)

Result

The bar chart was successfully created in R to show the distribution of employees across the departments, with the title, axes and data labels.

PROGRAM 3: Employee Performance Table

Experiment: Data Visualization using R

Question: Build a table to display the performance data for each employee. Label the table elements.

Aim

To build a table in R that displays the performance data of each employee, with the table elements labelled.

Procedure

1. Create the data frame with the columns Employee_ID, Department, Years_of_Service and Performance_Score.
2. Use cat() to display the table title Employee Performance Table.
3. Use print() with row.names = FALSE to display the table.

```
# Dataset
employees <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

# Table title
cat("Employee Performance Table\n\n")

# Display the table
print(employees, row.names = FALSE)
```

Table elements: the title is Employee Performance Table, the column headers are Employee_ID, Department, Years_of_Service and Performance_Score, and the data values are the entries in each employee row.

Expected Output

The table titled Employee Performance Table is displayed in the R console as shown below:

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

```
1 # Dataset
2 employees <- data.frame(
3   Employee_ID = c(1, 2, 3),
4   Department = c("Sales", "HR", "Marketing"),
5   Years_of_Service = c(5, 3, 7),
6   Performance_Score = c(85, 92, 78)
7 )
8
9 # Table title
10 cat("Employee Performance Table\n\n")
11
12 # Display the table
13 print(employees, row.names = FALSE)
14 |
```

14:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.5.0 ~/

```
> # Table title
> cat("Employee Performance Table\n\n")
Employee Performance Table

>
> # Display the table
> print(employees, row.names = FALSE)
Employee_ID Department Years_of_Service Performance_Score
      1      Sales          5             85
      2       HR          3             92
      3  Marketing          7             78

> |
```

Output

3: Employee Performance Table

Result

The employee performance table was successfully built in R, showing the department, years of service and performance score of each employee with the table title and headers labelled.

PROGRAM 4: Interactive Tableau Dashboard

Experiment: Data Visualization using Tableau

Question: Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

Aim

To develop an interactive Tableau dashboard that combines the line chart, the bar chart and the employee performance table for exploring the employee performance data.

Procedure

1. Save the Employee Performance dataset as an Excel (.xlsx) or CSV file. Open Tableau, click Microsoft Excel (or Text file) under Connect, select the file and click Open. Drag the sheet to the canvas.
2. Check the data types on the Data Source page: Employee ID, Years of Service and Performance Score must be numeric, and Department must be text. In the Data pane, right-click Employee ID and Years of Service and choose Convert to Dimension, so that they can be used as headers.
3. Create the line chart: click Sheet 1 and rename it Performance Trend. Drag Years of Service to Columns and Performance Score to Rows. Select Line in the Marks card, add circles, and turn on Show Mark Labels (Label > Show mark labels). Set the title to Employee Performance Trend and make sure the axis titles read Years of Service and Performance Score.
4. Create the bar chart: add a new worksheet and rename it Employees by Department. Drag Department to Columns and Number of Records (found at the bottom of the Data pane) to Rows. Select Bar in the Marks card and turn on Show Mark Labels. Set the title to Distribution of Employees Across Departments and the Y-axis title to Number of Employees.
5. Create the table: add a new worksheet and rename it Employee Performance Table. Drag Employee ID and Department to Rows, drag Measure Names to Columns, and drag Measure Values to Text in the Marks card. Keep only Years of Service and Performance Score in the Measure Values box. Set the title to Employee Performance Table.
6. Click the New Dashboard icon at the bottom of the screen. A blank dashboard opens with the list of worksheets on the left side.
7. Right-click the dashboard tab, choose Rename, and type Employee Performance Dashboard.
8. Set the dashboard size: in the left pane under Size, select Fixed size (for example, 1200 x 800) or Automatic so that all three components fit on one screen.
9. Drag the Performance Trend worksheet and drop it at the top-left part of the dashboard.
10. Drag the Employees by Department worksheet and drop it on the right side of the line chart (top-right part).
11. Drag the Employee Performance Table worksheet and drop it at the bottom of the dashboard, below both charts.
12. Arrange the components neatly by resizing their edges, so that all three are fully visible without overlapping.
13. Add the dashboard title: click Dashboard on the top menu and tick Show Title. Double-click the title and type Employee Performance Dashboard.
14. Add filters: go to each worksheet and drag Department to the Filters shelf (select all values and click OK). On the dashboard, click a chart, click the drop-down arrow at its top-right corner, choose Filters, and tick Department so that the filter box appears.
15. Apply the filter to all charts: click the drop-down arrow on the filter box, choose Apply to Worksheets, and select All Using This Data Source. Selecting a department (for example, HR) now updates the line chart, the bar chart and the table together.
16. Add interaction between the charts: click Dashboard on the top menu and choose Actions. Click Add Action, select Filter, set the Source Sheet as Employees by Department and the Target Sheets as Performance Trend and Employee Performance Table, set Run action on to Select, and click OK. (Alternatively, use the Use as Filter funnel icon on a chart.)

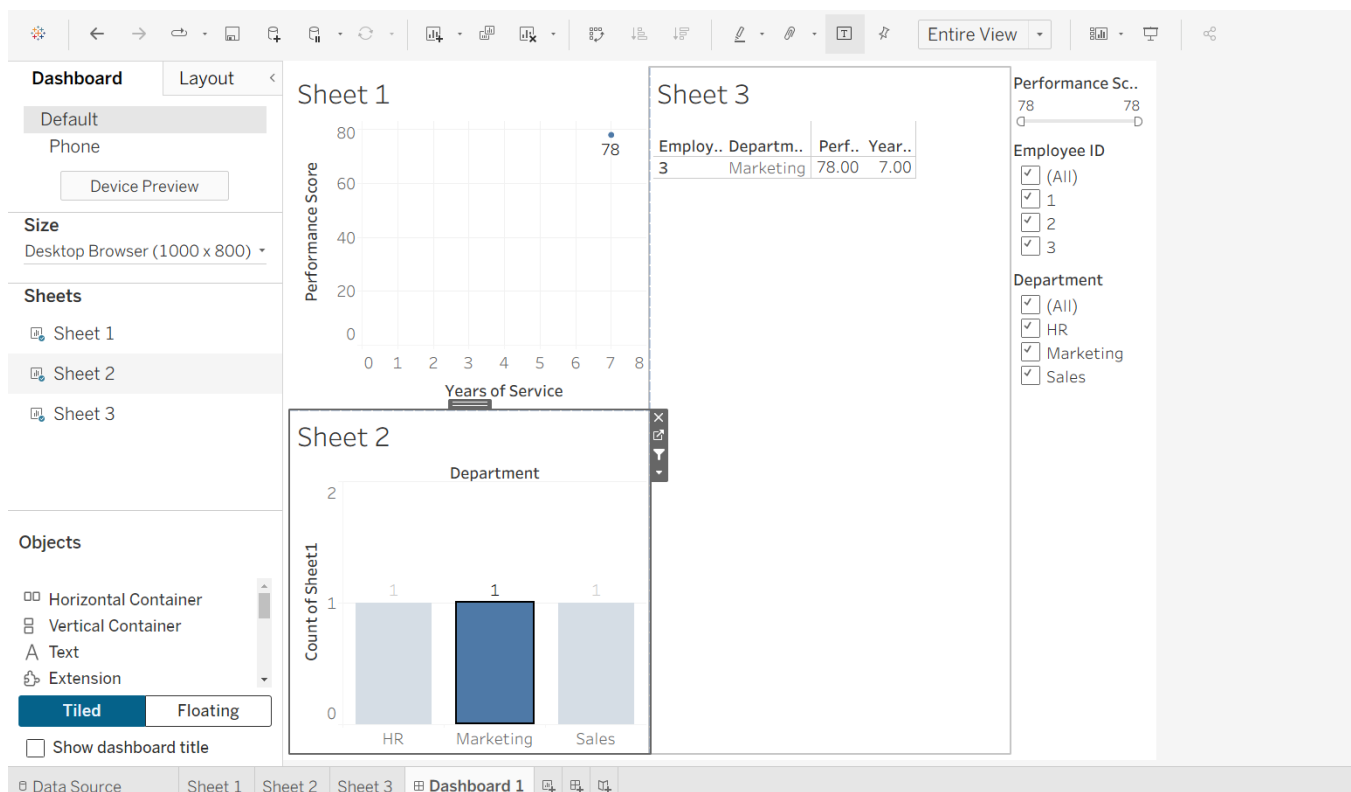
17. Test the dashboard by clicking a bar or selecting a department in the filter. The line chart and the table change to show only the selected department. Click a blank area (or choose All in the filter) to return to the full view.
18. Make the dashboard clean and readable by removing unnecessary legends and empty spaces, and keep the layout given below.

Dashboard Layout

- Top: Dashboard title – Employee Performance Dashboard
- Left/top: Line Chart
- Right/top: Bar Chart
- Bottom: Employee Performance Table
- Filter: Department

Expected Output

A dashboard titled Employee Performance Dashboard is displayed with the line chart on the top-left, the bar chart on the top-right and the employee performance table at the bottom, along with the Department filter. Selecting a department or a bar updates all three components for interactive exploration of the employee performance data.



Output 4: Employee Performance Dashboard

Result

The Employee Performance Dashboard was successfully created in Tableau by combining the line chart, the bar chart and the employee performance table, with a Department filter for interactive exploration of the employee performance data.